ಭಾರತೀಯ ಮಾಹಿತಿ ತಂತ್ರಜ್ಞಾನ ಸಂಸ್ಥೆ ರಾಯಚೂರು
भारतीय सूचना प्रौद्योगिकी संस्थान रायचूर
Indian Institute of Information Technology Raichur

CYBERSECURITY (AD310)

Assignment: Secure Drone Communication System

Total Marks: 10
Submission Mode: Online, Google Classroom. 2 PDFs.
Deadline: *29th April, 5 pm, through Google Classroom.*

---

1. Objective

The objective of this assignment is to design and implement a **secure communication system** by integrating multiple cryptographic concepts covered in the course, including:

- Symmetric Key Cryptography (AES)
- Asymmetric Cryptography (RSA, ECC)
- Diffie–Hellman Key Exchange
- Authentication Mechanisms
- Digital Signatures
- Message Integrity and Replay Protection

---

2. Problem Statement

You are required to design a **secure communication protocol** between a **Drone (Client)** and a **Ground Control Station (Server)**.

The system should ensure:

- Confidentiality of transmitted data
- Authentication of communicating entities
- Integrity of messages

- Protection against replay attacks

The implementation must simulate a **real-world drone communication scenario**, where the drone securely sends telemetry data (e.g., location, speed) to the ground station.

---

3. Tasks and Requirements

3.1 Key Exchange (2 Marks)

Implement a secure key exchange mechanism using either:

- Diffie–Hellman Key Exchange
  OR
- Elliptic Curve Cryptography (ECC)

Output: A shared secret key between the Drone and Ground Station.

3.2 Hybrid Encryption (2 Marks)

Implement a hybrid encryption scheme:

- Use RSA to securely exchange an AES session key
- Use AES (CBC or CTR mode) to encrypt the drone's data

Example Data Format:

```
{
  "drone_id": "DR001",
  "latitude": 12.97,
  "longitude": 77.59,
  "speed": 45
}
```

3.3 Authentication Mechanism (2 Marks)

Implement one of the following:

- Password-based authentication (with hashing and salting)

- Challenge–Response authentication
- Token-based authentication

Note:
Passwords must not be stored in plaintext. Use secure hash functions (e.g., SHA-256).

3.4 Digital Signature (1.5 Marks)

- Sign the message at the Drone side using RSA or ECC
- Verify the signature at the Ground Station

This ensures authenticity and non-repudiation.

3.5 Message Integrity (1.5 Marks)

- Generate a Message Authentication Code (MAC), such as HMAC-SHA256
- Verify the MAC at the receiver side

3.6 Replay Attack Protection (1 Mark)

Implement a mechanism to prevent replay attacks using:

- Timestamp
  OR
- Nonce

Repeated or outdated messages must be rejected.

---

4. Expected Workflow

1. Drone authenticates with the Ground Station
2. Key exchange is performed (DH/ECC)
3. AES session key is securely shared (via RSA or derived key)
4. Drone:
    - Encrypts data using AES

    - Signs the message
    - Attaches MAC and timestamp/nonce
5. Ground Station:
    - Verifies authentication
    - Verifies digital signature
    - Validates MAC
    - Checks replay protection
    - Decrypts the message

---

5. Implementation Guidelines

- Programming Language: **Python**
- Allowed Libraries:
    - pycryptodome
    - cryptography
- Code should be modular, for example:
    - key_exchange.py
    - encryption.py
    - authentication.py
    - main.py

---

6. Evaluation Criteria (10 Marks)

Component	Marks
Key Exchange	2
Hybrid Encryption	2
Authentication	2
Digital Signature	1.5
Message Integrity	1.5
Replay Protection	1
Total	10

---

7. Submission Requirements

Students must submit:

1. PDF-1:Screenshots of the experiments (filename as output_roll_number.pdf)
    a. **Output Screenshots** demonstrating execution
2. PDF-2: Code followed by an explanation of the code. (filename as code_roll_number.pdf
    a. **Python Code** (well-structured and commented)
    b. **Short Report (Max 2 Pages)** including:
        - System design
        - Explanation of implementation
        - Security analysis:
            - What attacks are prevented?
            - What are the limitations?

---

8. Bonus (Optional – +1 Mark)

Simulate and demonstrate one of the following attacks:

- Replay Attack
- Man-in-the-Middle Attack

Show how your system detects or prevents the attack.

---

9. Academic Integrity

- This assignment must be completed individually
- *You may use any source, any tool, and do the experiment by yourself, but not copy from friends. FYI, I have subscribed to tools to evaluate plagiarism. And just looking at the output screenshots, I would be able to judge plagiarism.*
- Plagiarism or copying will result in zero marks